SOFTWARE REQUIREMENTS SPECIFICATION

Domain: Digital Governance and E-Services

Problem Statement: Unified Citizen Service Delivery Platform (UCSDP)

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1. Preface

Expected Readership

This Software Requirements Specification (SRS) document has been prepared for everyone involved in bringing the Unified Citizen Service Delivery Platform (UCSDP) to life—from faculty reviewers and project stakeholders to system analysts, designers, developers, and quality assurance teams working within the digital governance and e-services space. We've written this document in straightforward, accessible language to ensure that readers from all backgrounds, whether deeply technical or not, can grasp the system's requirements and vision.

Summary

Since this is our first version, there aren't any changes from previous iterations to report. Moving forward, we anticipate that future versions will reflect updates driven by stakeholder input, shifts in government policies, emerging technologies, or natural evolution in what the system needs to accomplish.

2. Introduction

2.1. Need for the System

Right now, government services are scattered across numerous departments and platforms, which creates real headaches for everyday citizens trying to get things done. People face long wait times, lack of clarity about where their applications stand, the frustration of submitting the same documents multiple times, and the inconvenience of having to physically show up at government offices just to complete simple tasks.

The Unified Citizen Service Delivery Platform (UCSDP) exists to solve these problems. We're building a single digital hub where citizens can access a wide range of government services in a way that's convenient, transparent, and well-organized—all from the comfort of their homes or local service centers.

2.2. System Overview and Functions

The UCSDP serves as a centralized digital platform designed to bring essential government services online. The system accommodates various user types and delivers services including:

- Applying for and renewing official certificates (such as income, caste, and residential certificates) entirely online
- Managing welfare programs, subsidies, and scholarships with built-in eligibility verification
- Handling permissions and authorizations for public events and vendor activities
- Updating and correcting official records
- Submitting and resolving grievances or complaints
- Tracking application progress, scheduling appointments, and receiving timely notifications

The platform ensures smooth interactions between citizens and government personnel through well-defined workflows and role-specific access controls.

2.3. Interaction with Other Systems

The UCSDP won't work in isolation—it will coordinate with various supporting systems to deliver services effectively. These supporting systems include:

- Citizen record and identity databases for verification purposes
- Payment interfaces to handle fee collection (in a simulated environment for now)

These connections will enable accurate identity verification, prompt communication, and consistent data handling across all services offered.

2.4. Business and Strategic Objectives

The UCSDP directly supports the digital governance goals of the organization commissioning it. Specifically, the system aims to:

- Make government services more accessible and efficient for everyone
- Cut down on manual paperwork and eliminate administrative bottlenecks
- Increase transparency and accountability in how services are delivered
- Foster a more citizen-focused and inclusive approach to governance

By meeting these objectives, the UCSDP plays a vital role in the broader digital transformation journey and helps improve how public services are managed over the long term.

3. Glossary

Here are the key terms used throughout this document:

Term	Definition
Unified Citizen Service Delivery Platform (UCSDP)	A web-based system that brings together multiple government services on a single online platform, making them easily accessible to citizens.
Digital Governance	The practice of using digital technologies by government organizations to deliver services and engage with citizens more effectively.

E-Services	Government services provided electronically through the internet, eliminating the need for in-person visits to physical offices.
Citizen	An individual who uses the system to submit applications for government services or to report issues and complaints.
Department Officer (First-Level Officer)	A government official tasked with reviewing and processing citizen applications as the first point of official evaluation.
Department Supervisor (Second-Level Officer)	A senior official who oversees service processing, manages escalated situations, and has the authority to override decisions when necessary.
Grievance	A formal complaint or concern raised by a citizen regarding service delays, application rejections, payment issues, or overall service quality.
Grievance Officer	An official specifically designated to receive, investigate, and work toward resolving grievances submitted by citizens.
Service Facilitator	An individual or service center (such as a Common Service Center or MeeSeva center) that helps citizens navigate and use government services.
Authentication	The process of confirming a user's identity before granting them access to the system.
SLA (Service Level Agreement)	A specified time frame within which a particular service or grievance must be processed and completed.

4. User Requirements Definition

This section outlines the services and capabilities that the system will provide to each type of user.

4.1 User Services (Functional Requirements)

4.1.1 Citizen

- The system will enable citizens to submit applications for various government services and certificates.
- The system will allow citizens to upload and manage the documents they need to support their applications.
- The system will let citizens monitor the status of their applications in real-time.
- The system will provide citizens with a way to raise grievances or complaints related to any service they've received.

- The system will enable citizens to request corrections to their official records and schedule appointments with relevant departments.

4.1.2 Department Officer (First-Level Officer)

- The system will allow officers to carefully review and verify applications that have been submitted.
- The system will give officers the ability to approve or reject applications based on their assessment.
- The system will enable officers to request additional information or clarifications from citizens when needed.

4.1.3 Department Supervisor (Second-Level Officer)

- The system will allow supervisors to keep track of pending and overdue applications across their department.
- The system will enable supervisors to step in and manage escalated cases, including the ability to override decisions when circumstances warrant it.

4.1.4 Grievance Officer

- The system will allow grievance officers to receive and begin processing citizen complaints.
- The system will enable grievance officers to resolve complaints directly or escalate them to higher authorities when necessary.

4.1.5 Service Facilitator

- The system will allow facilitators to submit applications on behalf of citizens who may need assistance.
- The system will enable facilitators to help with document uploads and assist citizens in filing grievances.

4.2. Non-Functional User Requirements

- The system will be intuitive and user-friendly, requiring minimal training for people to get started.
- The system will respond to user actions promptly, keeping wait times within reasonable limits.
- The system will maintain secure handling and storage of all user data.
- The system will remain accessible to users around the clock, with exceptions only for scheduled maintenance periods.

4.3. Product and Process Standards

- The system will adhere to recognized best practices for web application development.
- The system will comply with fundamental digital governance guidelines established for public services.
- User data will be managed in accordance with prevailing data privacy and security standards.

5. System Requirements Specification

5.1 Functional Requirements

- **5.1.1** The system will provide a secure registration and login process for all users.
- **5.1.2** The system will enable both citizens and service facilitators to submit service applications.
- **5.1.3** The system will support uploading, verifying, and securely storing documents.
- **5.1.4** The system will give department officers the ability to approve, reject, or ask for additional information regarding applications.
- **5.1.5** The system will facilitate the complete grievance lifecycle, including registration, processing, escalation, and closure.
- **5.1.6** The system will track the status of applications and grievances according to pre-defined workflows.
- **5.1.7** The system will allow supervisors to oversee and intervene in escalated cases as needed.

5.2 Non-Functional Requirements

- **5.2.1 Performance:** The system will deliver responses to user actions within timeframes that users find acceptable.
- **5.2.2 Security:** The system will guarantee secure access controls and robust protection of user data.
- **5.2.3 Availability:** The system will be accessible to users at all times, with brief interruptions only during scheduled maintenance windows.
- **5.2.4 Reliability:** The system will manage failures gracefully and maintain data consistency even under adverse conditions.
- **5.2.5 Usability:** The system will feature an intuitive design that requires minimal training for effective use.

5.3 External Interfaces

- **5.3.1 User Interface:** The system will offer a web-based interface that works seamlessly with standard web browsers.

- **5.3.2 Notification Interface:** The system will connect with email and SMS services to keep users informed about important updates.
- **5.3.3 Database Interface:** The system will interact with a centralized database to store and retrieve all relevant system data efficiently.

6. System Models

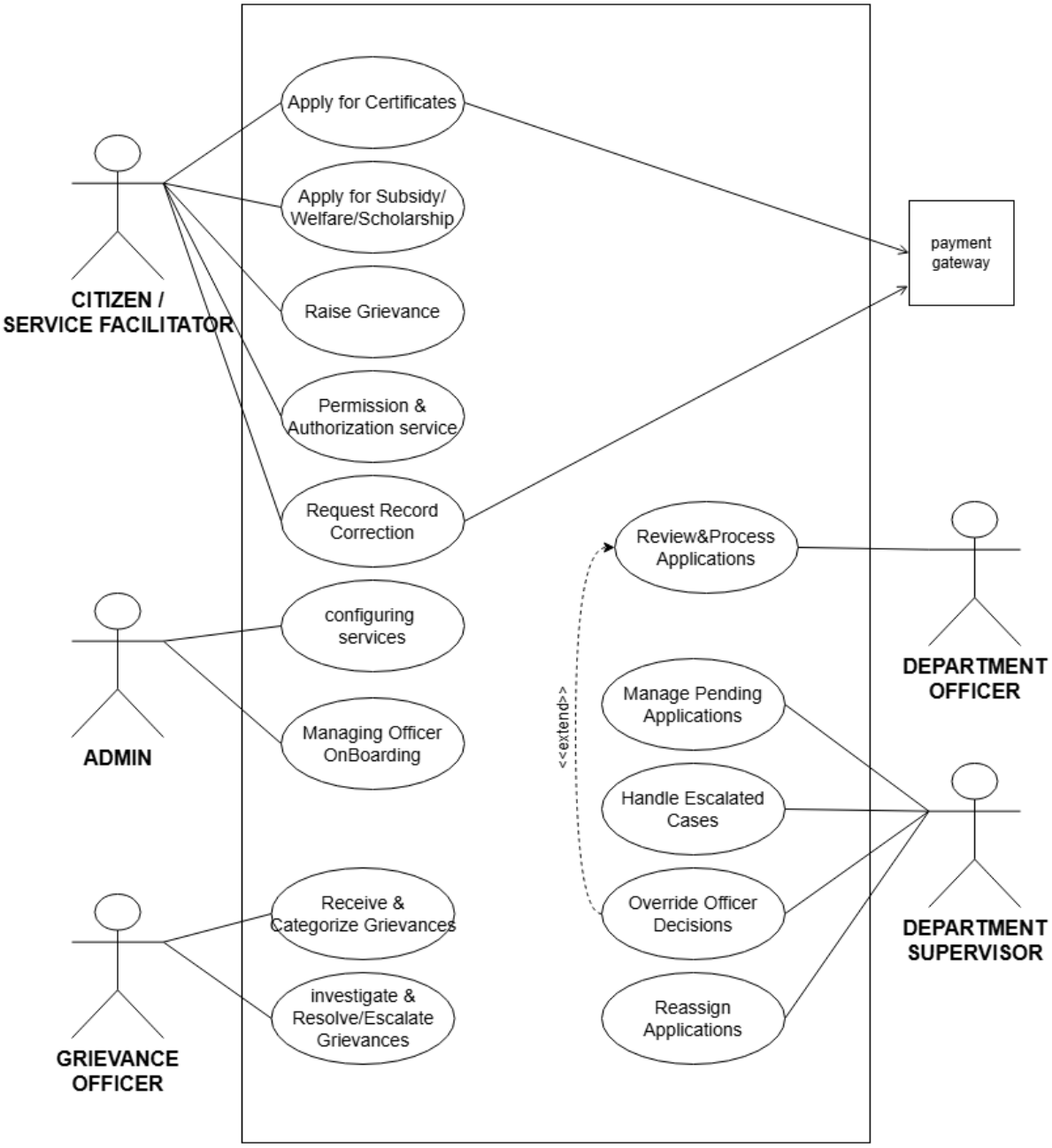
This section includes various diagrams and models that help visualize how the system functions:

- Use case diagrams
- Activity diagram
- Sequence diagrams

Use case diagram

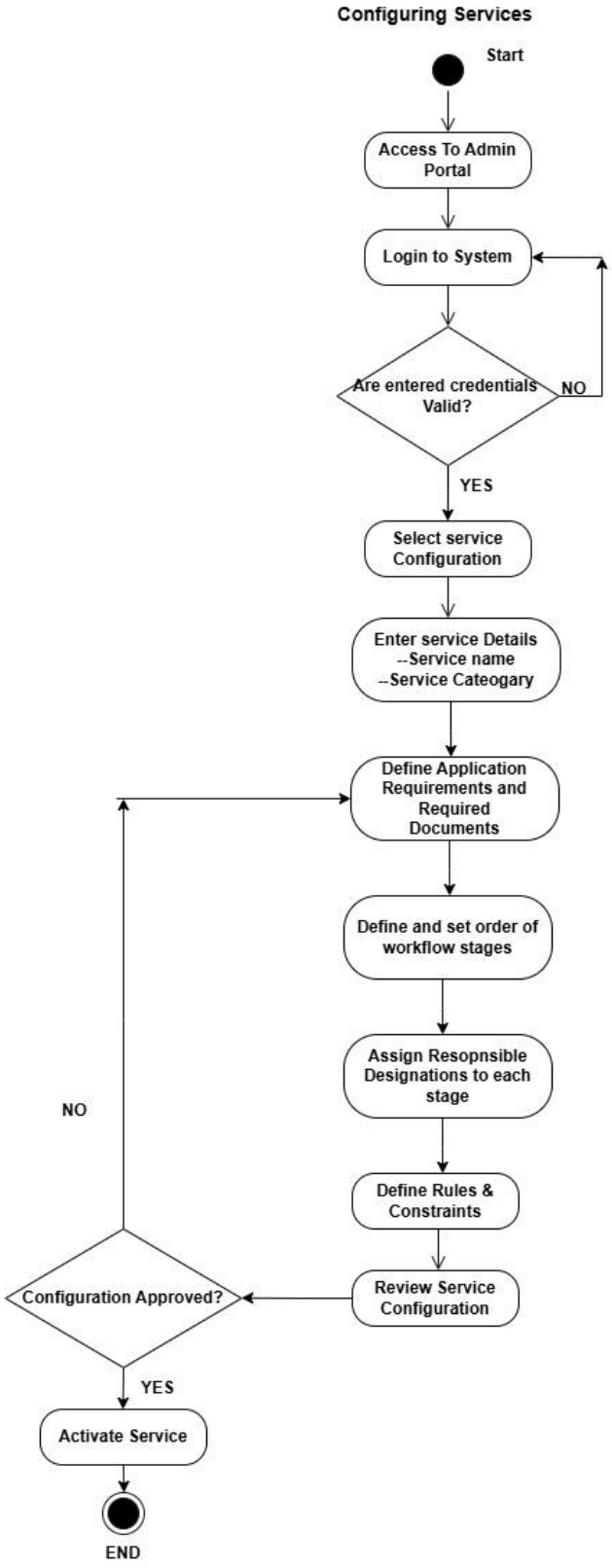
UNIFIED CITIZEN SERVICE DELIVERY PLATFORM

E-Governance System

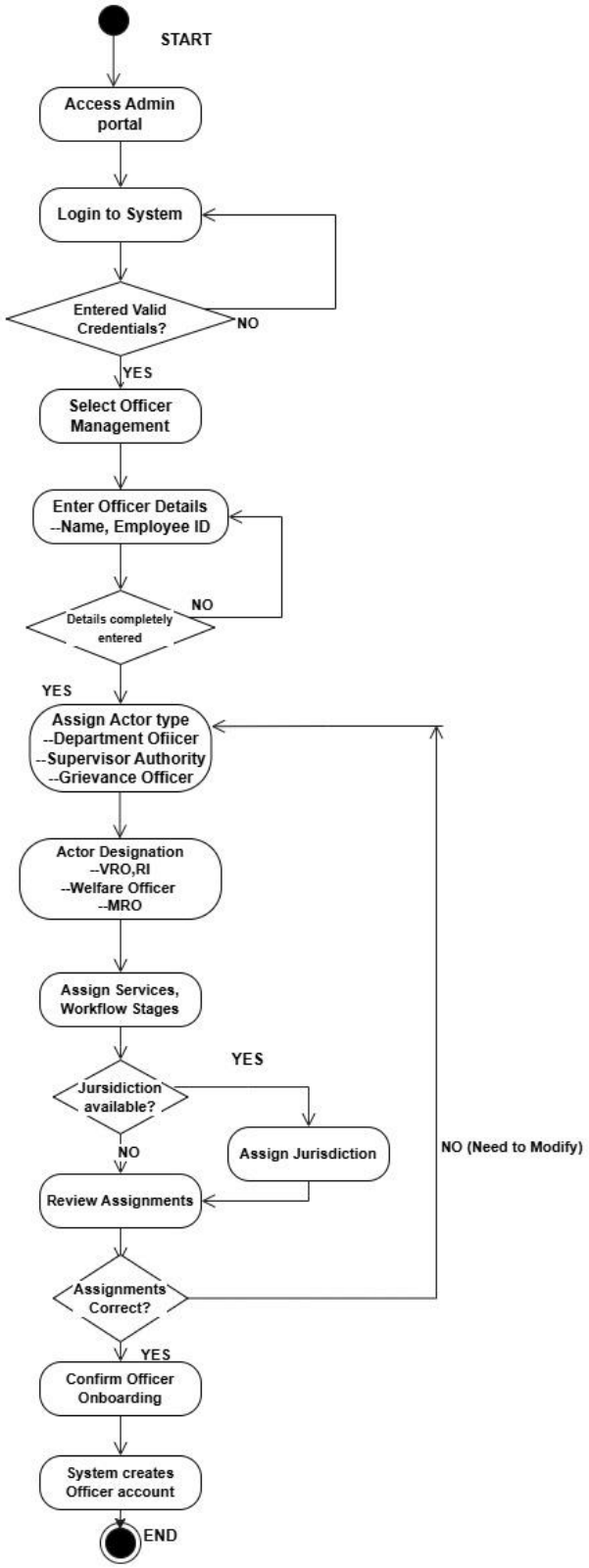


Activity diagrams

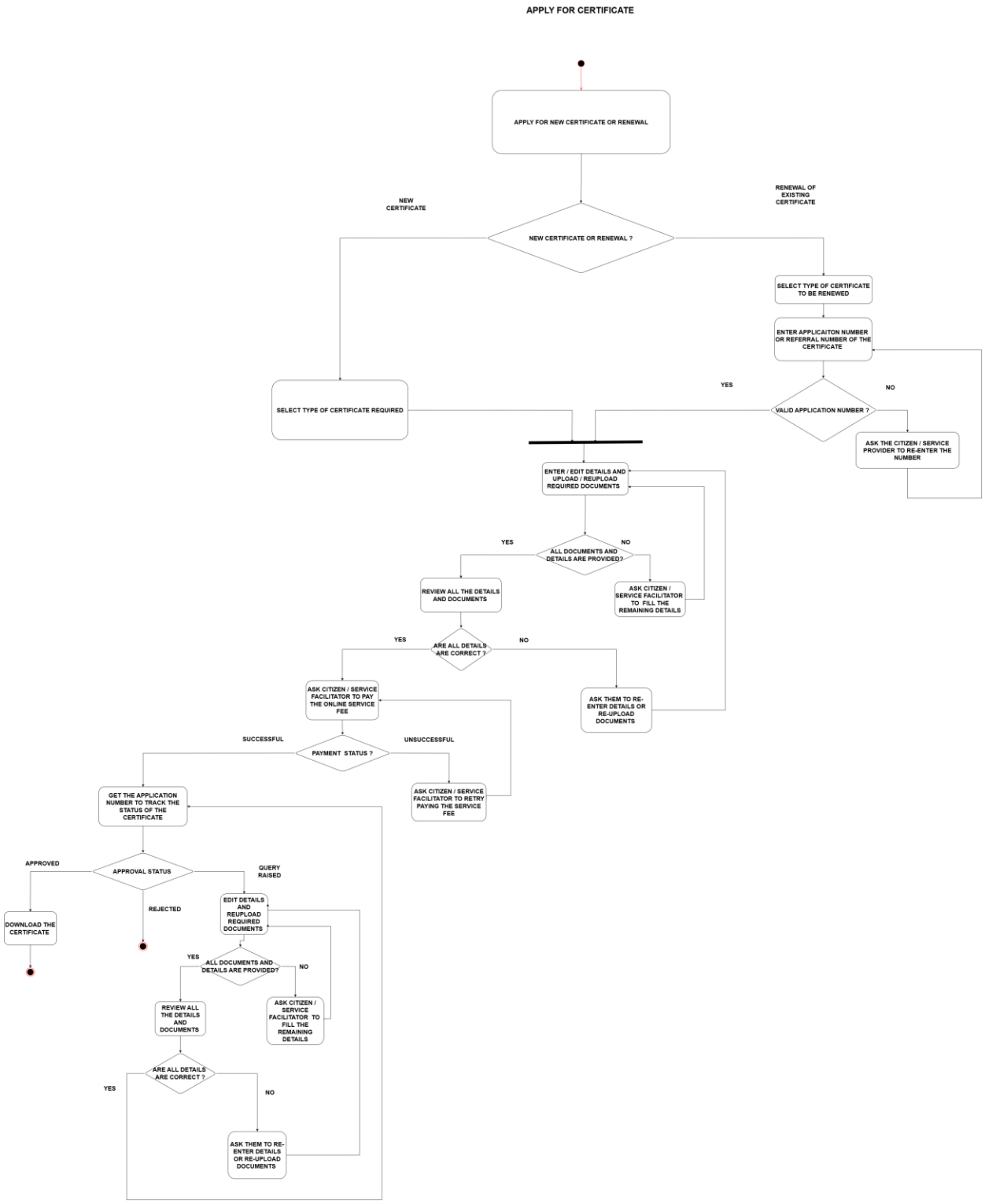
1.Admin


Configuring Services
Start
Access To Admin Portal
Login to System
Are entered credentials Valid?
NO
YES
Select service Configuration
Enter service Details
--Service name
--Service Cateogary
Define Application Requirements and Required Documents
Define and set order of workflow stages
Assign Resopnsible Designations to each stage
Define Rules & Constraints
Review Service Configuration
Configuration Approved?
NO
YES
Activate Service
END

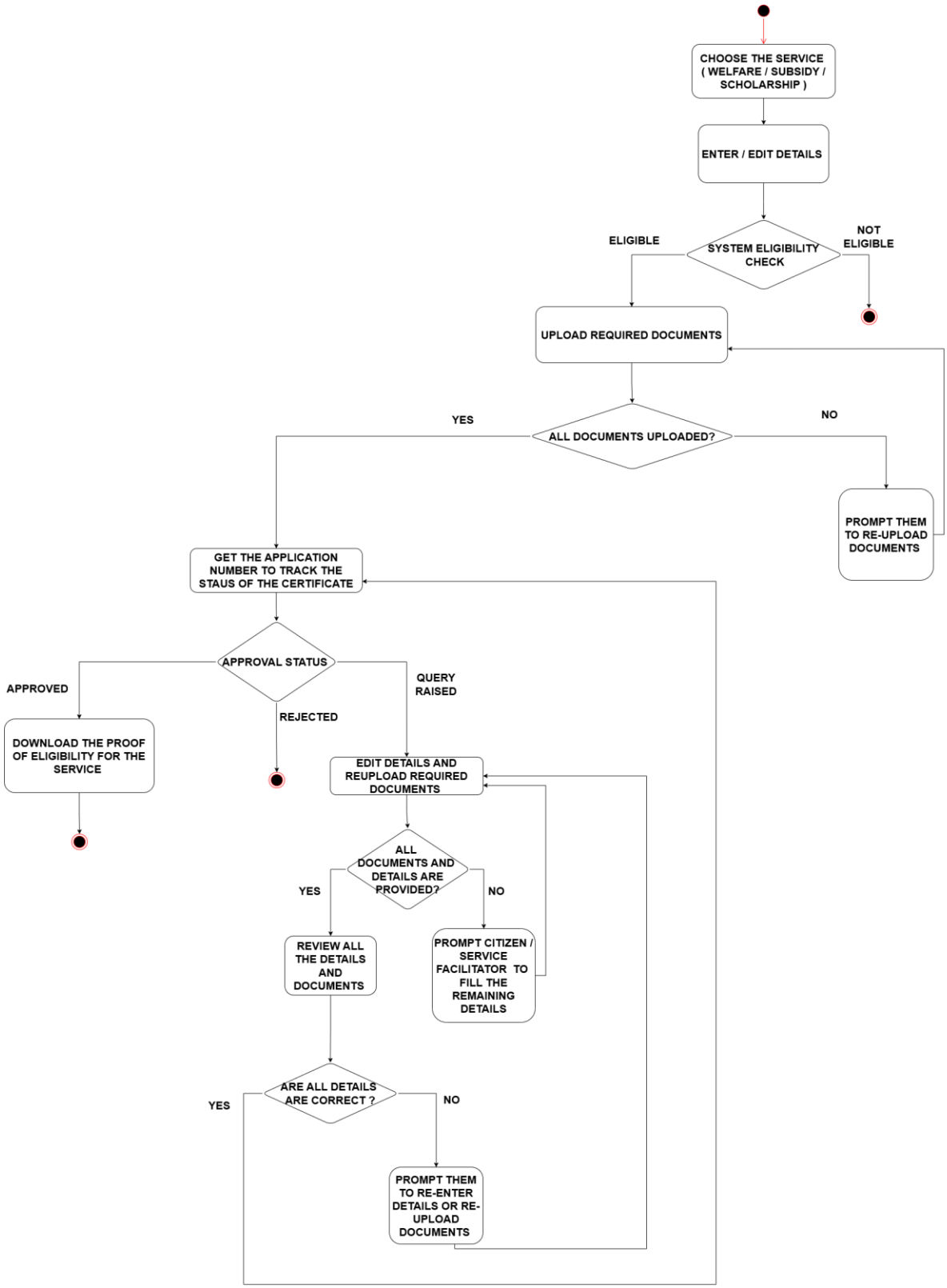
MANAGE OFFICER ONBOARDING



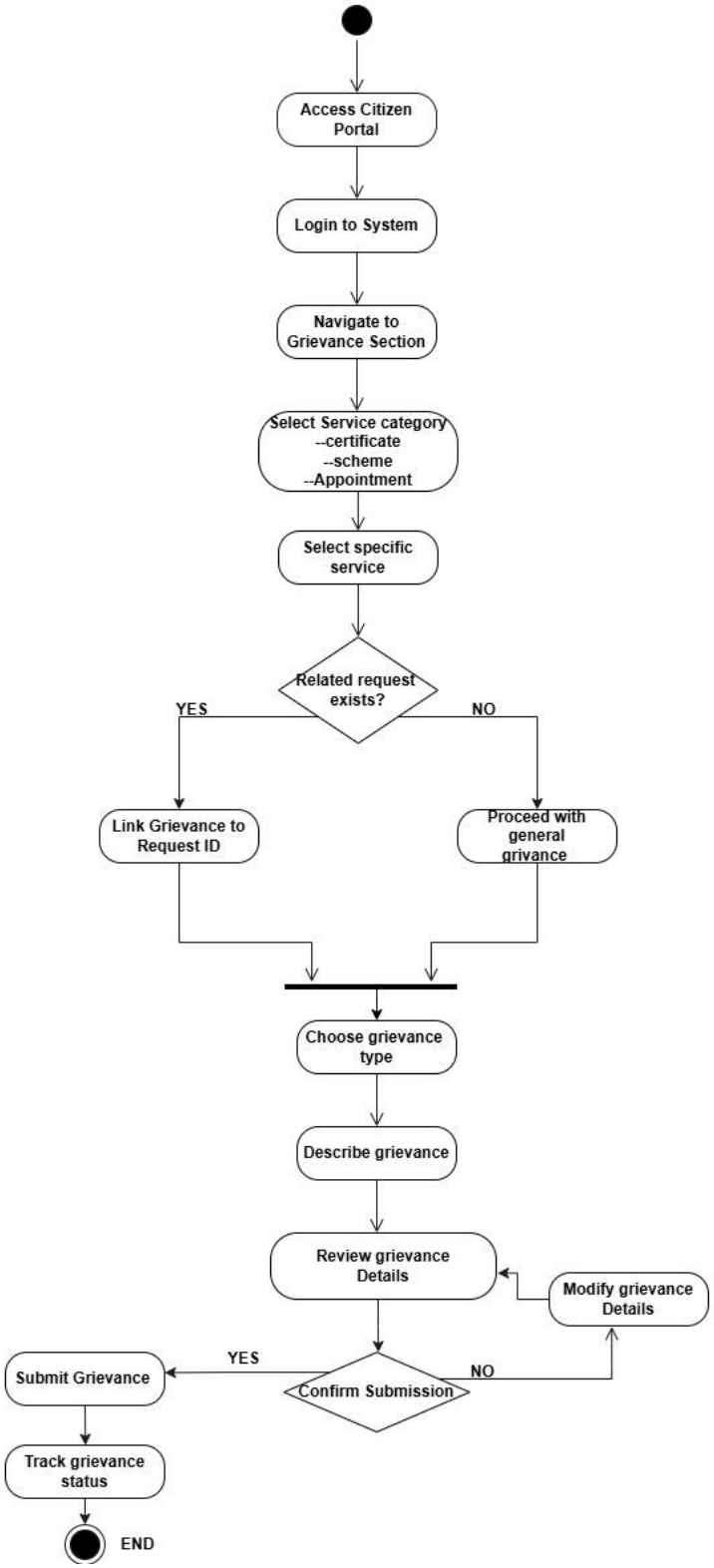
2.Citizen


APPLY FOR CERTIFICATE
APPLY FOR NEW CERTIFICATE OR RENEWAL
NEW CERTIFICATE
RENEWAL OF EXISTING CERTIFICATE
NEW CERTIFICATE OR RENEWAL ?
SELECT TYPE OF CERTIFICATE TO BE RENEWED
ENTER APPLICAITON NUMBER OR REFERRAL NUMBER OF THE CERTIFICATE
YES
NO
VALID APPLICATION NUMBER ?
ASK THE CITIZEN / SERVICE PROVIDER TO RE-ENTER THE NUMBER
SELECT TYPE OF CERTIFICATE REQUIRED
ENTER / EDIT DETAILS AND UPLOAD / REUPLOAD REQUIRED DOCUMENTS
YES
NO
ALL DOCUMENTS AND DETAILS ARE PROVIDED?
REVIEW ALL THE DETAILS AND DOCUMENTS
ASK CITIZEN / SERVICE FACILITATOR TO FILL THE REMAINING DETAILS
YES
NO
ARE ALL DETAILS ARE CORRECT ?
ASK CITIZEN / SERVICE FACILITATOR TO PAY THE ONLINE SERVICE FEE
ASK THEM TO RE-ENTER DETAILS OR RE-UPLOAD DOCUMENTS
SUCCESSFUL
UNSUCCESSFUL
PAYMENT STATUS ?
ASK CITIZEN / SERVICE FACILITATOR TO RETRY PAYING THE SERVICE FEE
GET THE APPLICATION NUMBER TO TRACK THE STATUS OF THE CERTIFICATE
APPROVED
APPROVAL STATUS
QUERY RAISED
REJECTED
DOWNLOAD THE CERTIFICATE
EDIT DETAILS AND REUPLOAD REQUIRED DOCUMENTS
YES
ALL DOCUMENTS AND DETAILS ARE PROVIDED?
NO
REVIEW ALL THE DETAILS AND DOCUMENTS
ASK CITIZEN / SERVICE FACILITATOR TO FILL THE REMAINING DETAILS
ARE ALL DETAILS ARE CORRECT ?
YES
NO
ASK THEM TO RE-ENTER DETAILS OR RE-UPLOAD DOCUMENTS

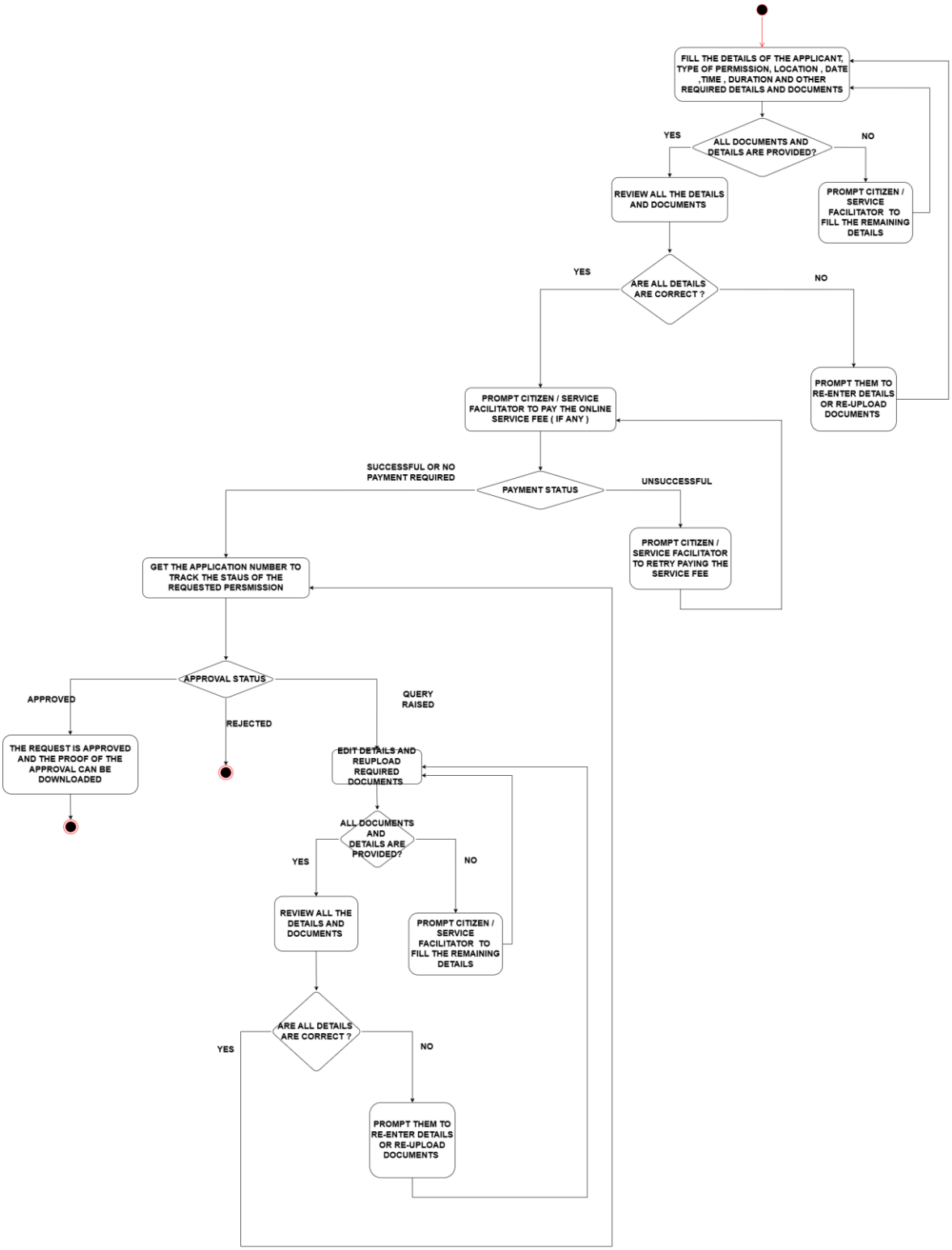
APPLY FOR SUBSIDY / WELFARE / SCHORLARSHIP



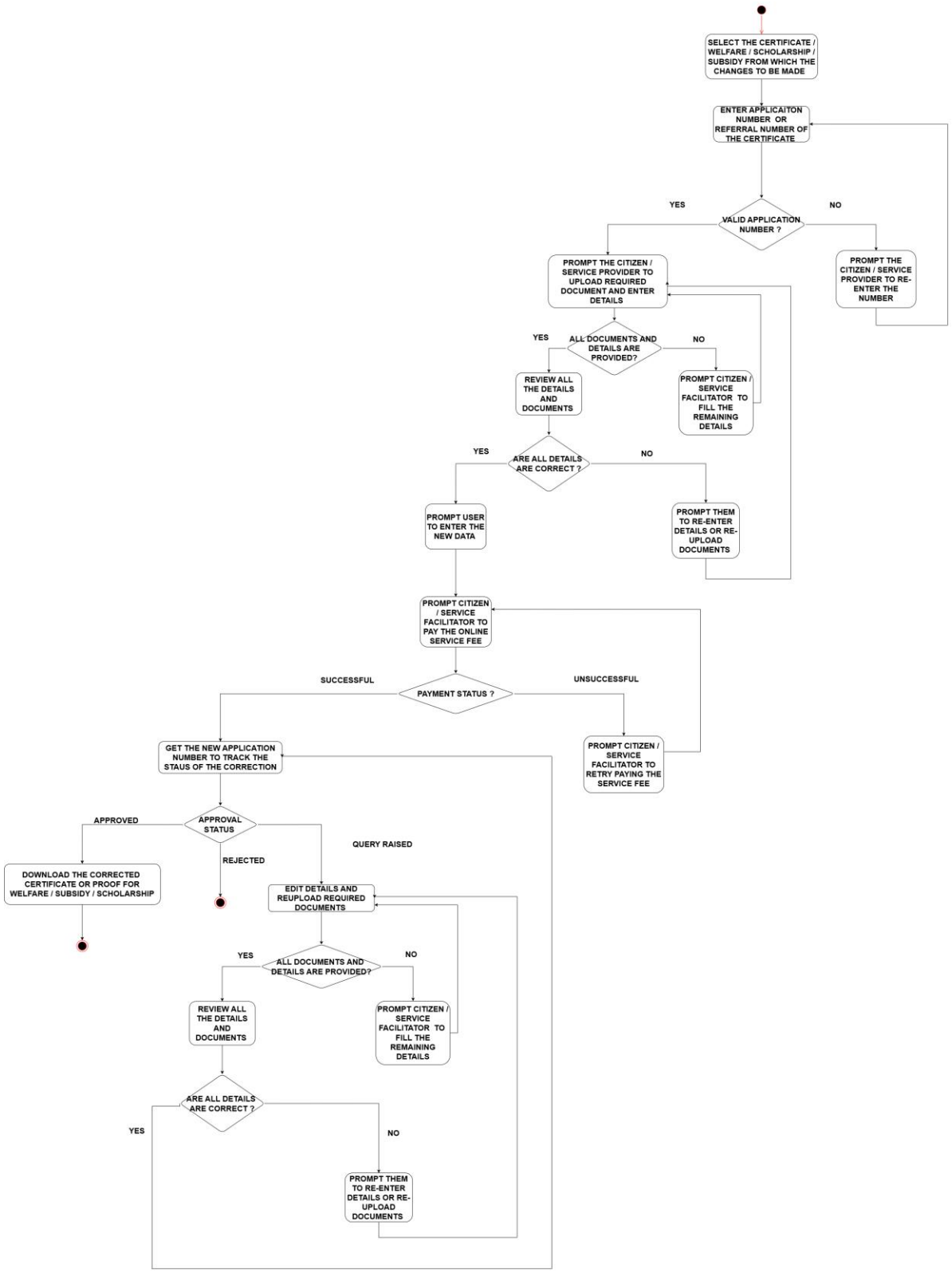
RAISE GRIEVANCE



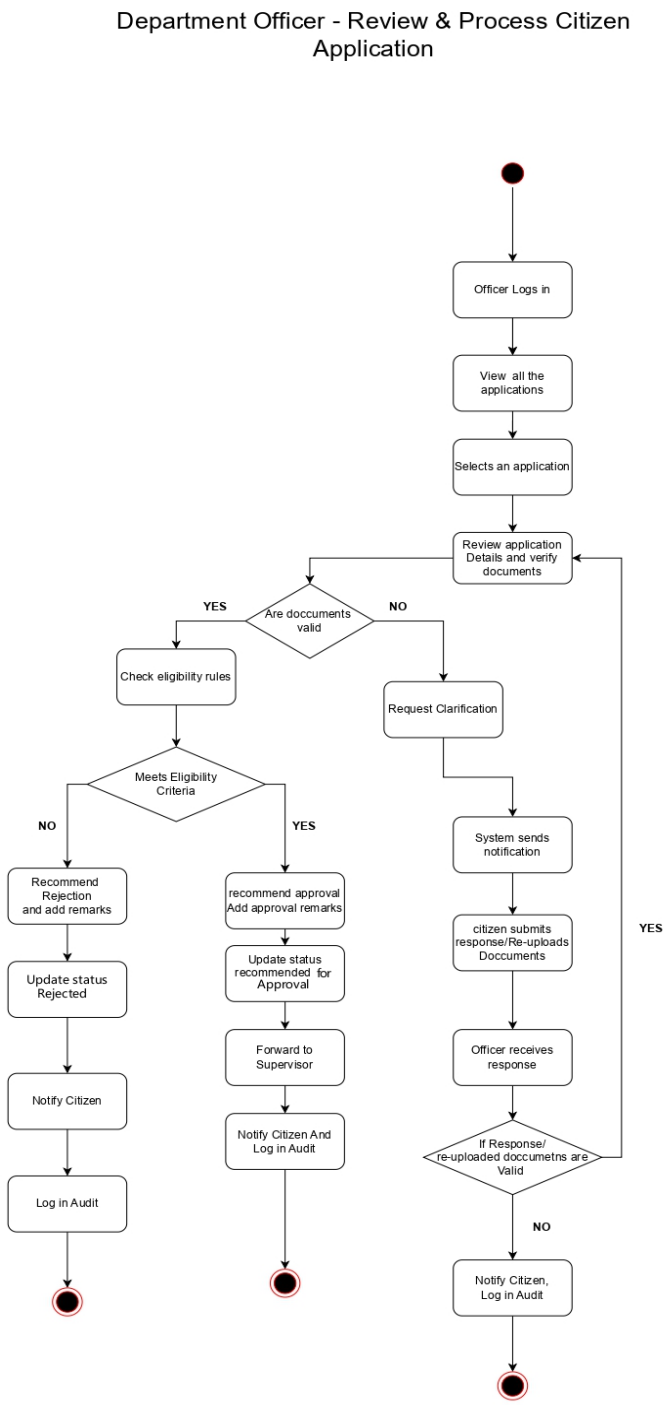
PERMISSION AND AUTHORIZATION SERVICE



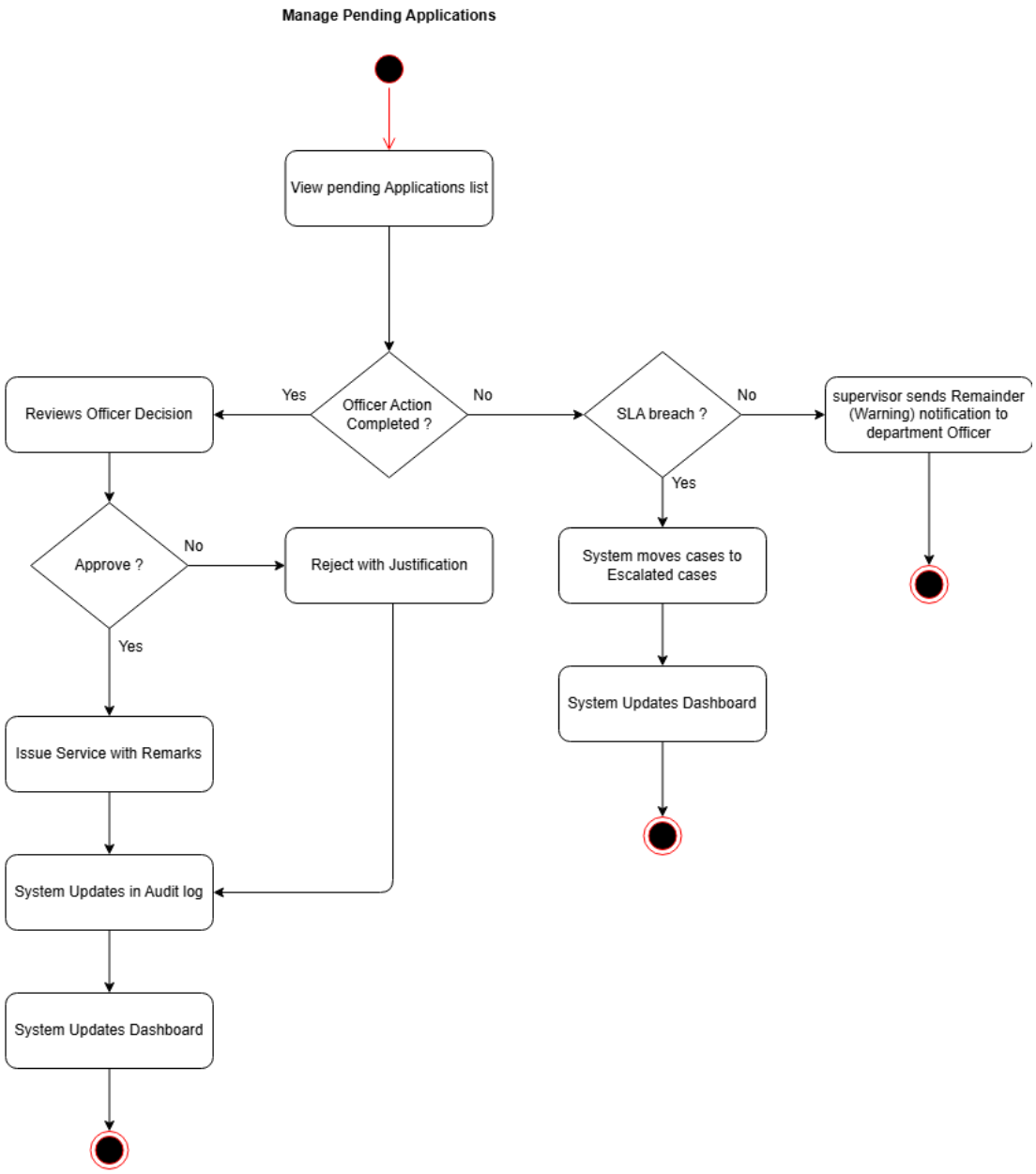
REQUEST RECORD CORRECTION

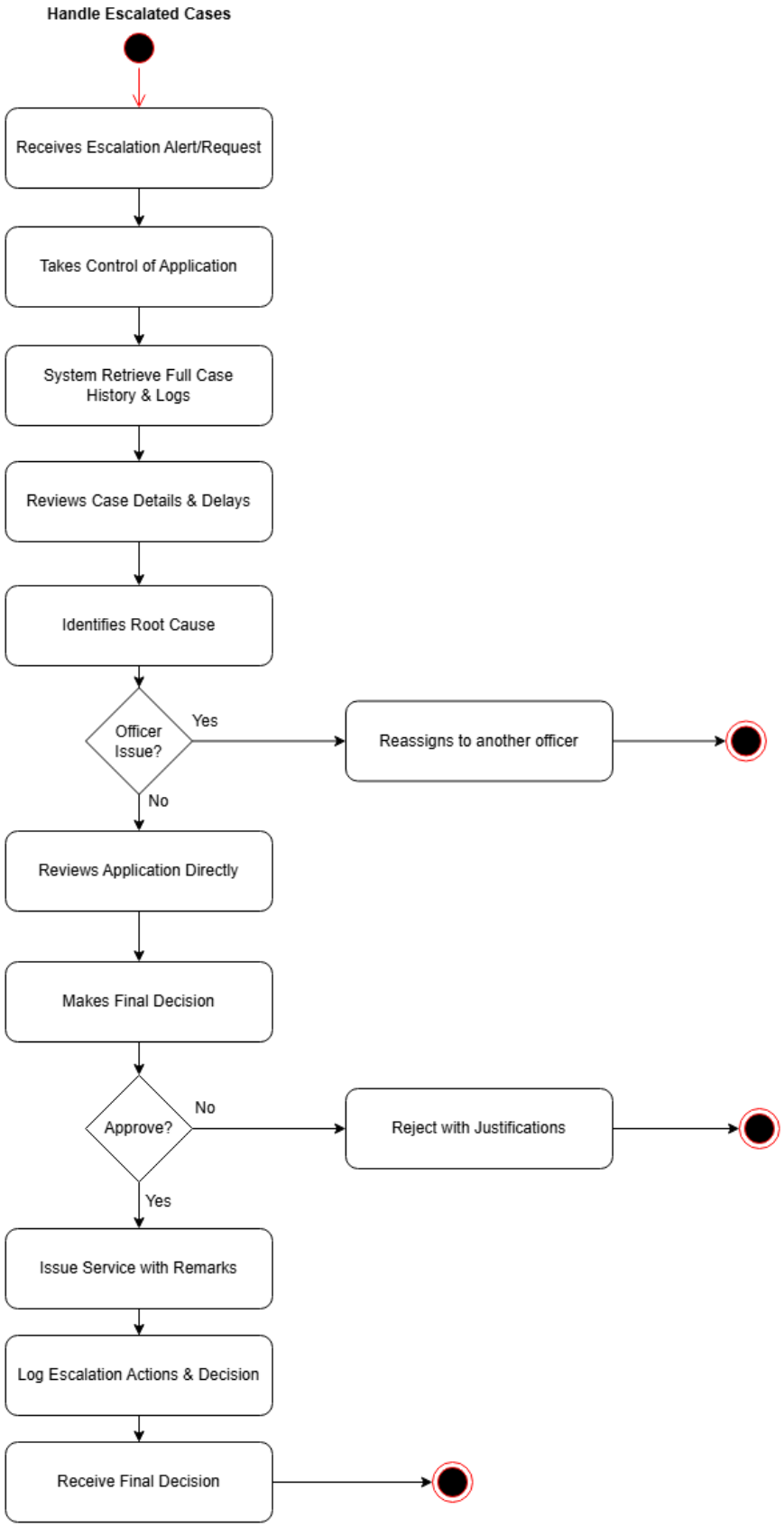


3.Department Officer

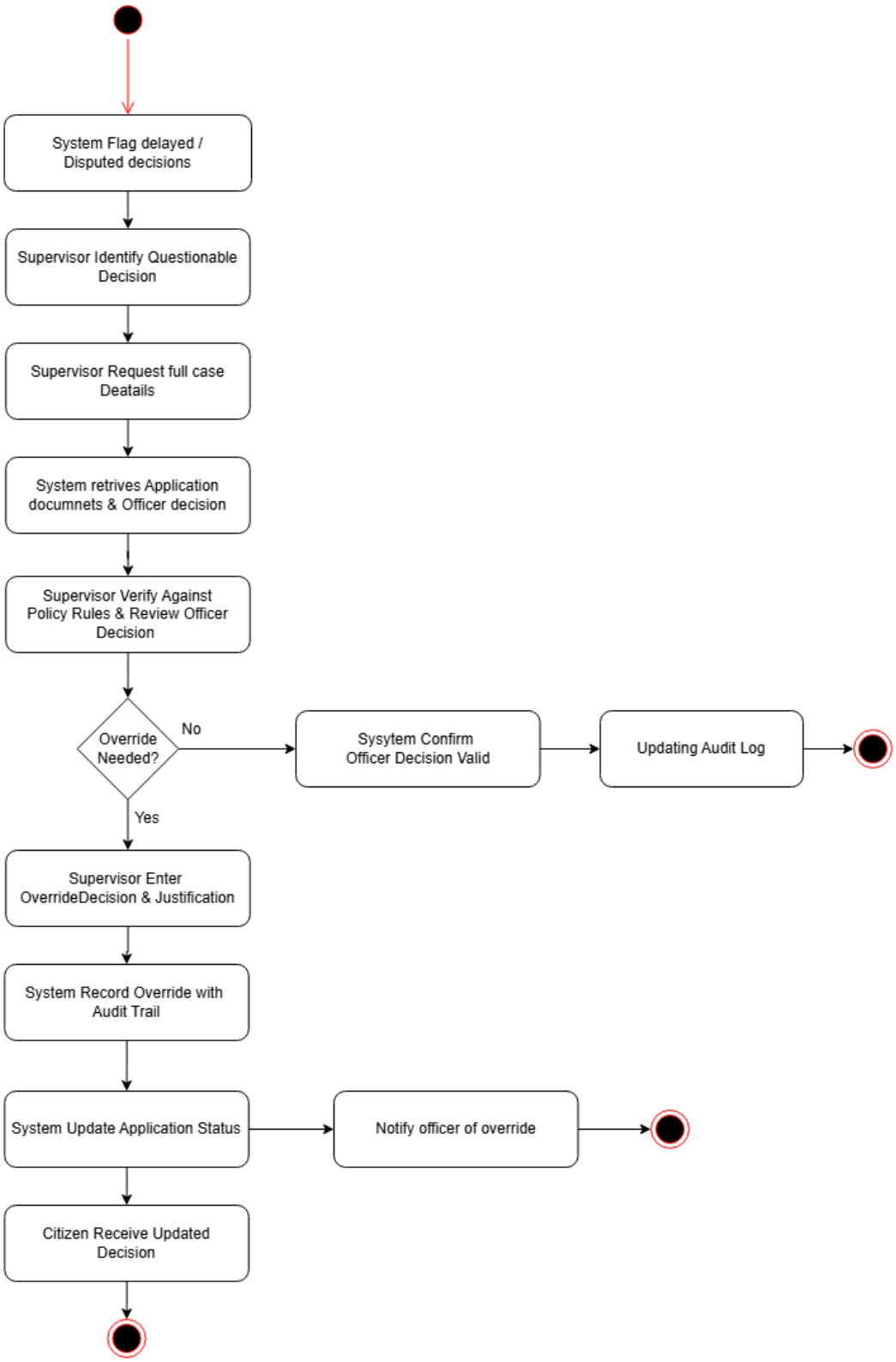


4.Department Supervisor

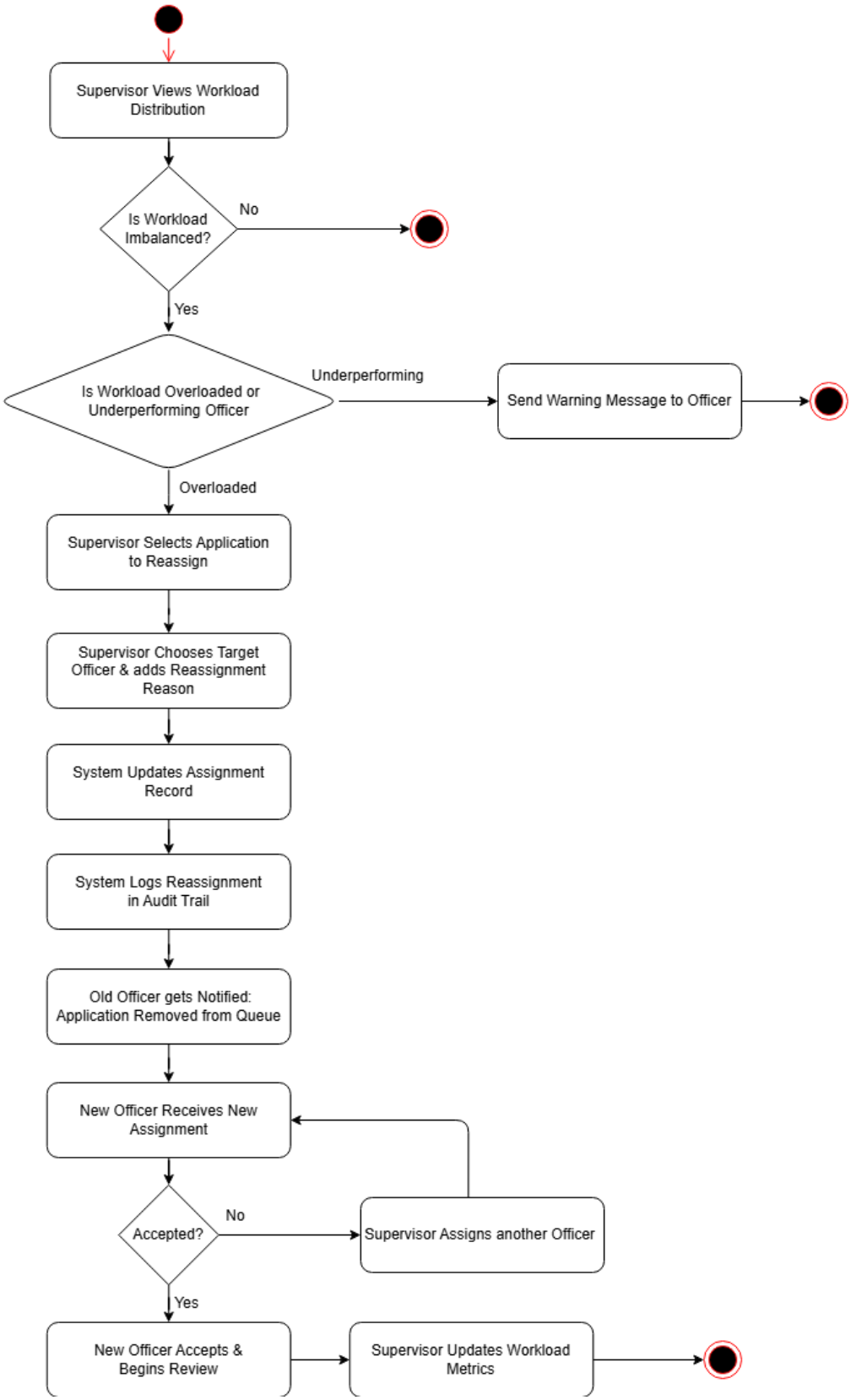



Handle Escalated Cases
Receives Escalation Alert/Request
Takes Control of Application
System Retrieve Full Case History & Logs
Reviews Case Details & Delays
Identifies Root Cause
Officer Issue?
Yes
Reassigns to another officer
No
Reviews Application Directly
Makes Final Decision
Approve?
No
Reject with Justifications
Yes
Issue Service with Remarks
Log Escalation Actions & Decision
Receive Final Decision

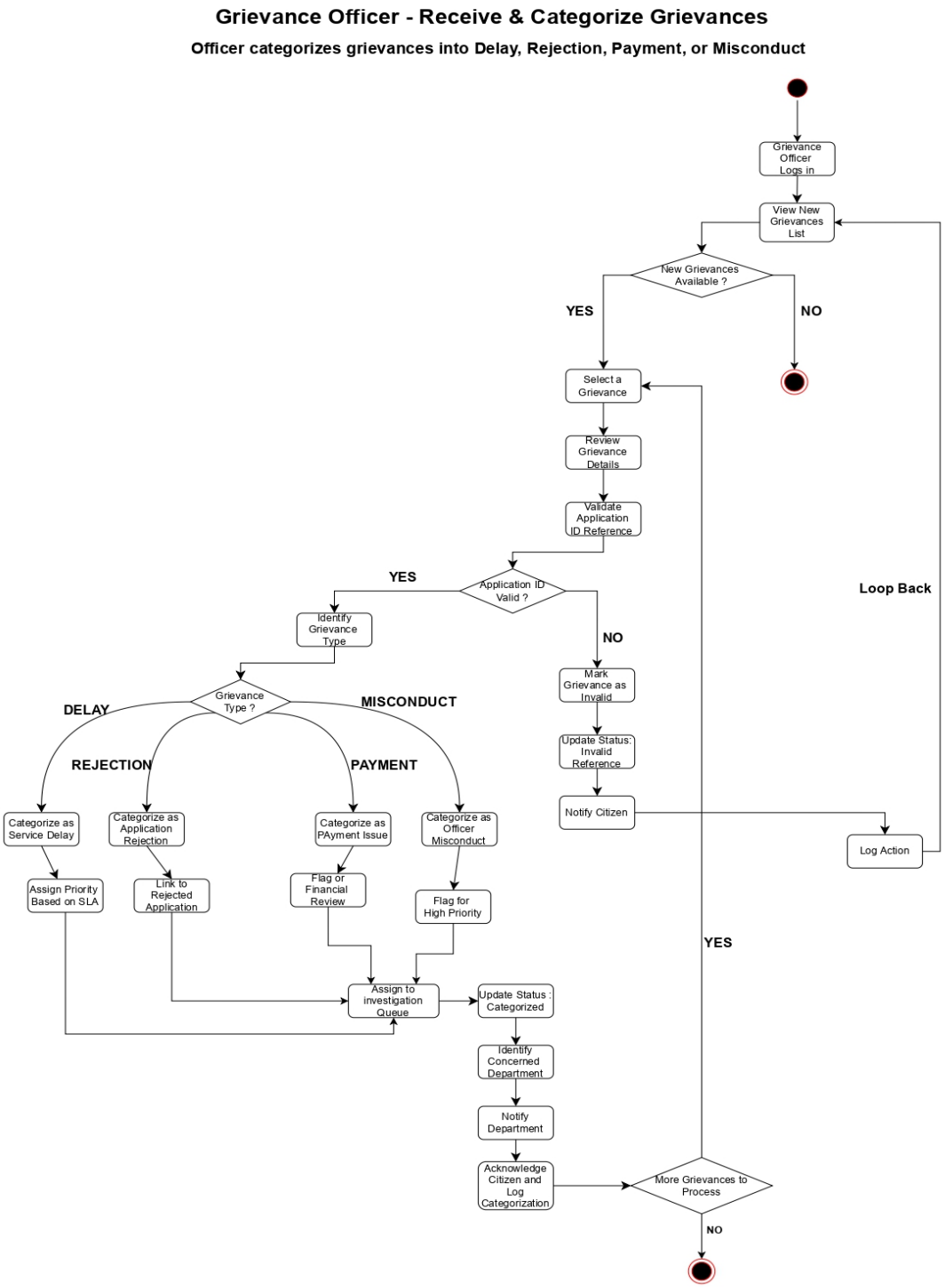
Override Officer Decisions



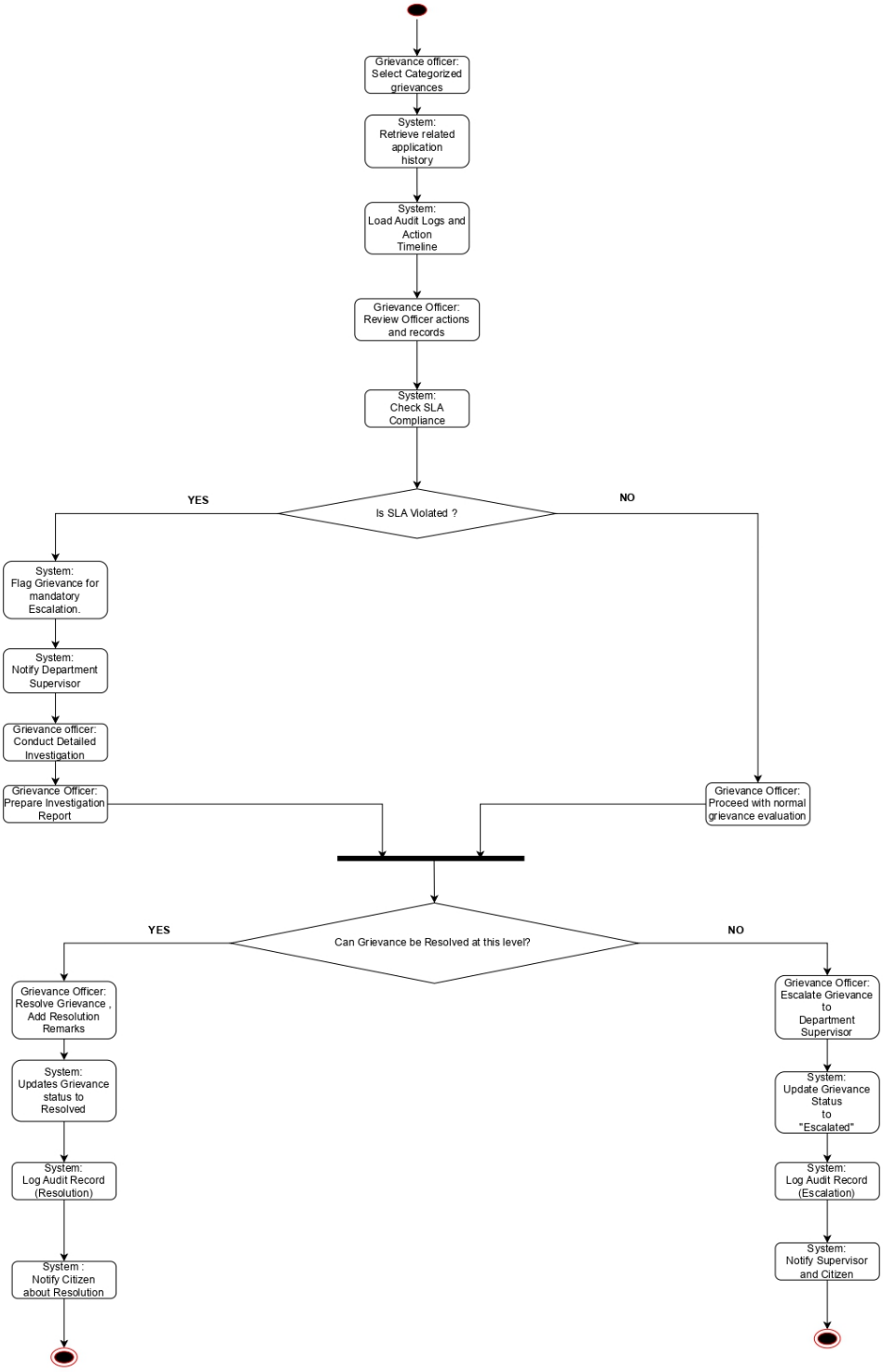
Reassign Applications



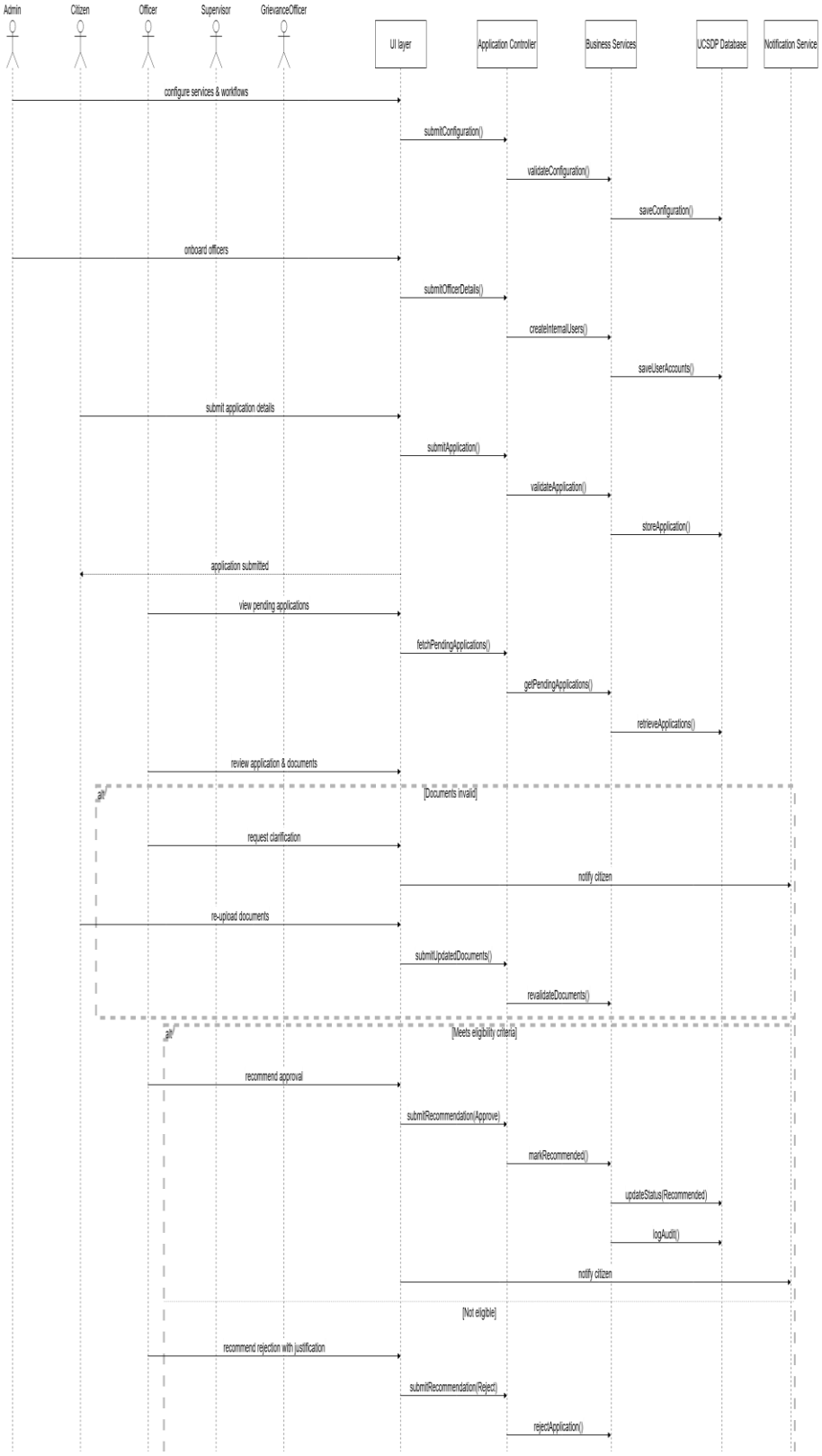
5. Grievance Officer

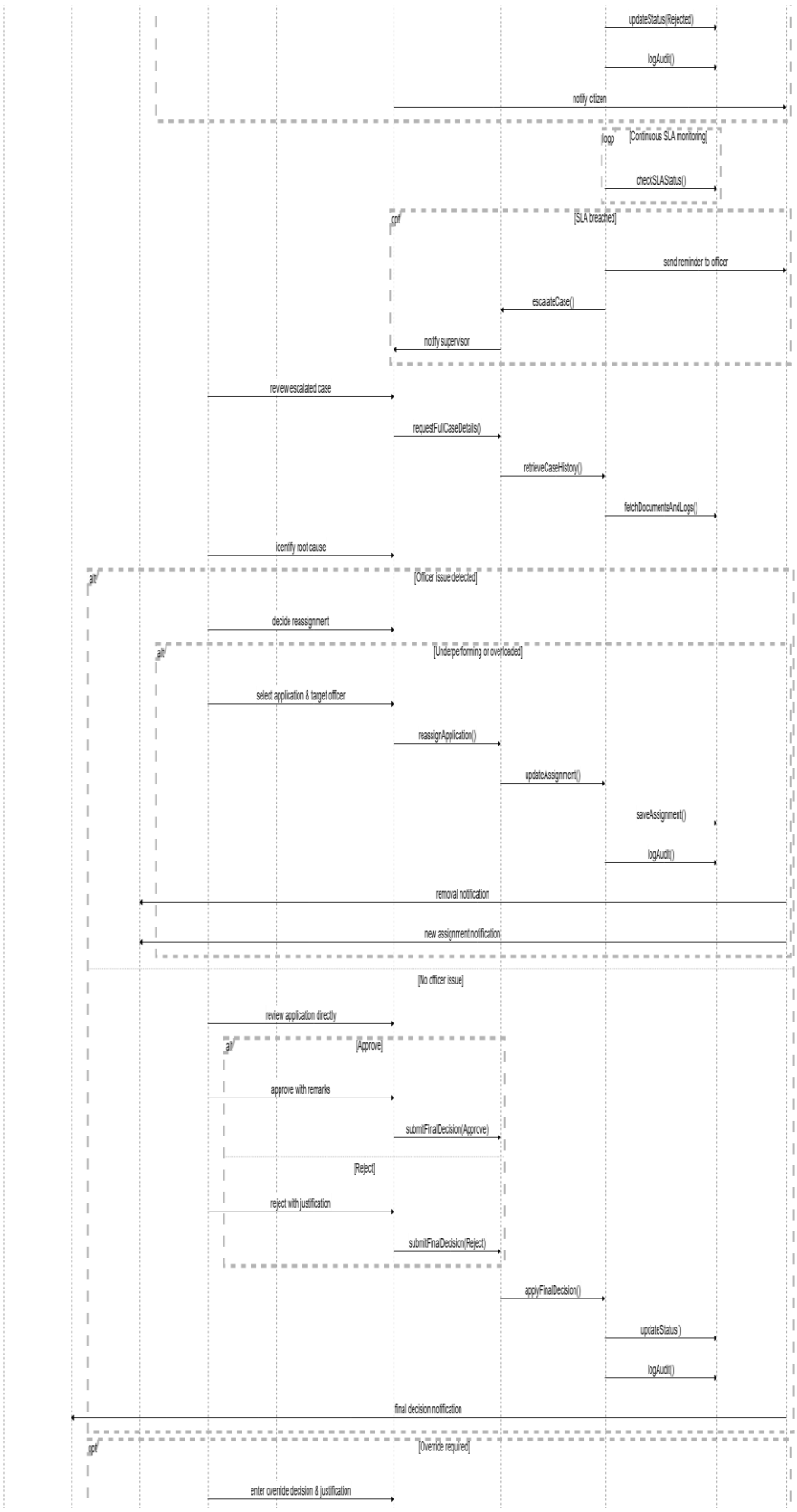


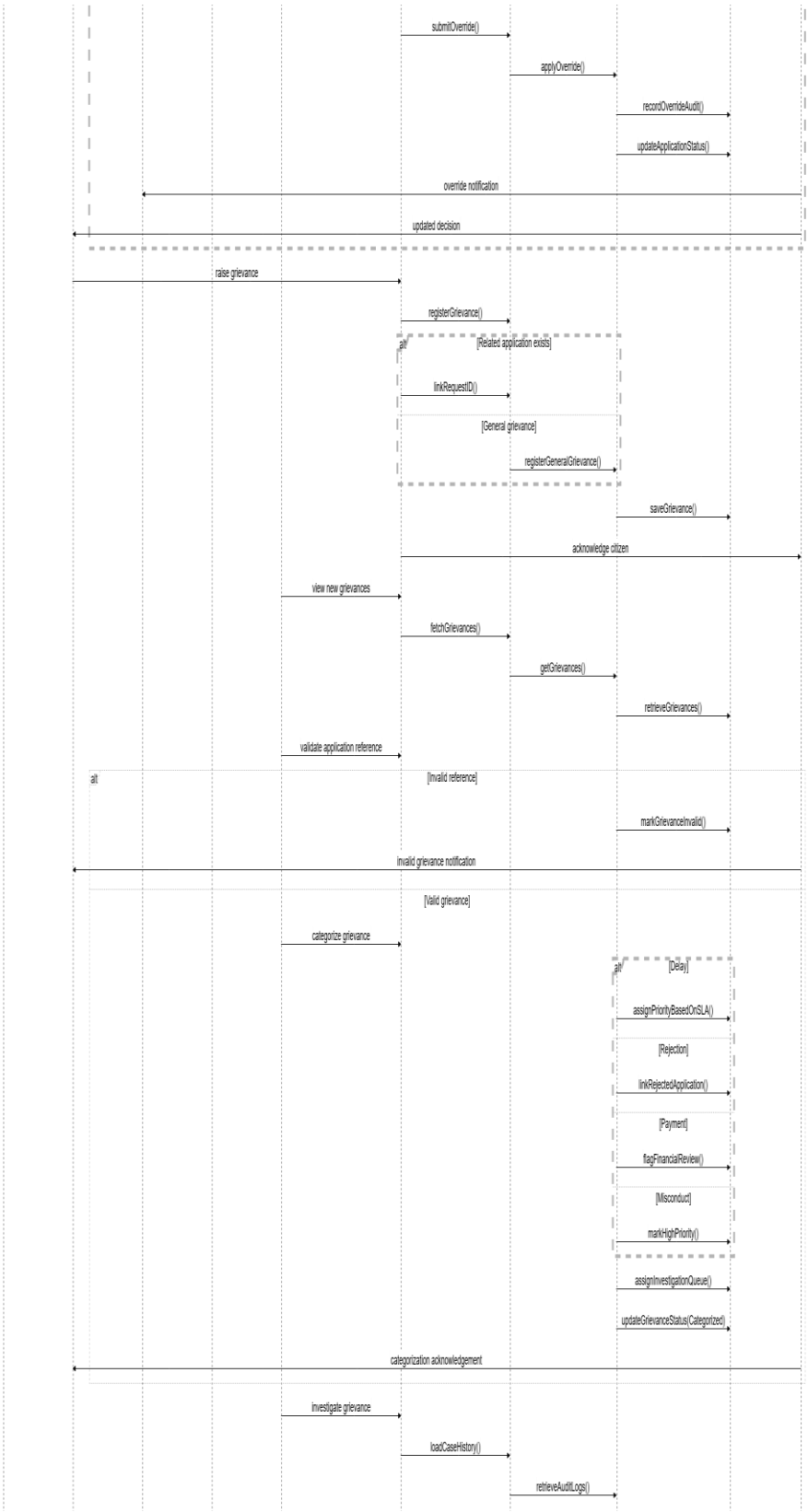
Grievance Officer – Investigate & Resolve / Escalate Grievance

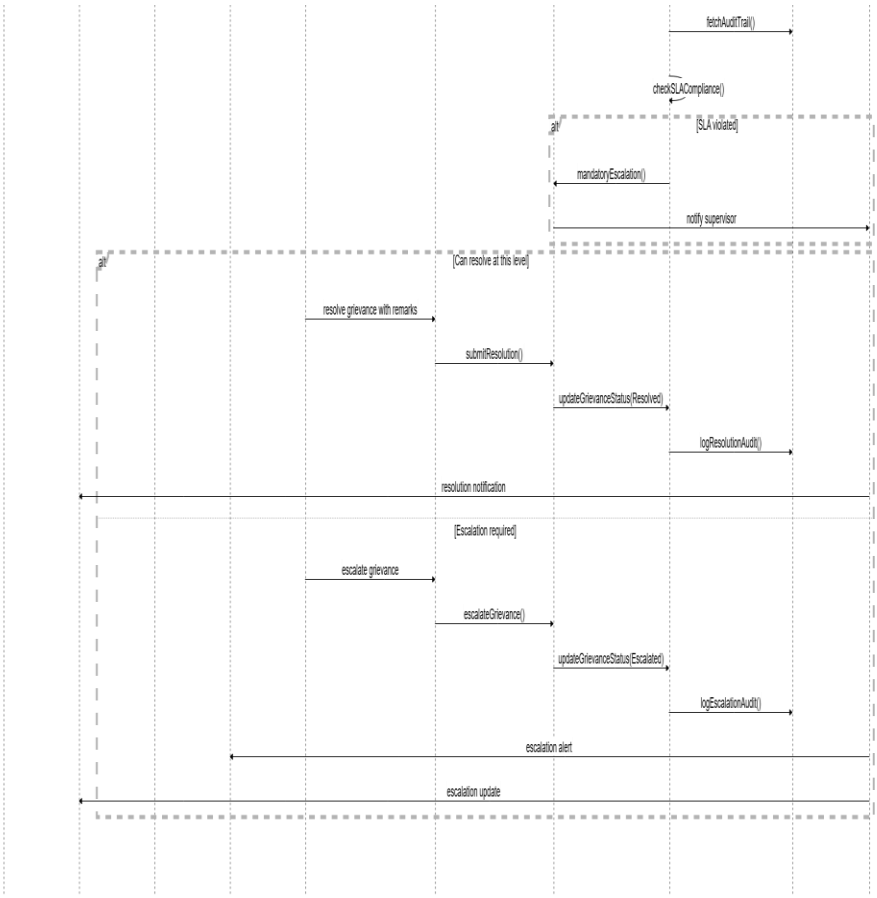


Sequence diagram


Admin
Citizen
Officer
Supervisor
GrievanceOfficer
UI layer
Application Controller
Business Services
UCSDP Database
Notification Service
configure services & workflows
submitConfiguration()
validateConfiguration()
saveConfiguration()
onboard officers
submitOfficerDetails()
createInternalUsers()
saveUserAccounts()
submit application details
submitApplication()
validateApplication()
storeApplication()
application submitted
view pending applications
fetchPendingApplications()
getPendingApplications()
retrieveApplications()
review application & documents
alt
[Documents invalid]
request clarification
notify citizen
re-upload documents
submitUpdatedDocuments()
revalidateDocuments()
alt
[Meets eligibility criteria]
recommend approval
submitRecommendation(Approve)
markRecommended()
updateStatus(Recommended)
logAudit()
notify citizen
[Not eligible]
recommend rejection with justification
submitRecommendation(Reject)
rejectApplication()


updateStatus(Rejected)
logAudit()
notify citizen
loop
[Continuous SLA monitoring]
checkSLAStatus()
opt
[SLA breached]
send reminder to officer
escalateCase()
notify supervisor
review escalated case
requestFullCaseDetails()
retrieveCaseHistory()
fetchDocumentsAndLogs()
identify root cause
alt
[Officer issue detected]
decide reassignment
alt
[Underperforming or overloaded]
select application & target officer
reassignApplication()
updateAssignment()
saveAssignment()
logAudit()
removal notification
new assignment notification
[No officer issue]
review application directly
alt
[Approve]
approve with remarks
submitFinalDecision(Approve)
[Reject]
reject with justification
submitFinalDecision(Reject)
applyFinalDecision()
updateStatus()
logAudit()
final decision notification
opt
[Override required]
enter override decision & justification


submitOverride()
applyOverride()
recordOverrideAudit()
updateApplicationStatus()
override notification
updated decision
raise grievance
registerGrievance()
alt
[Related application exists]
linkRequestID()
[General grievance]
registerGeneralGrievance()
saveGrievance()
acknowledge citizen
view new grievances
fetchGrievances()
getGrievances()
retrieveGrievances()
validate application reference
alt
[Invalid reference]
markGrievanceInvalid()
invalid grievance notification
[Valid grievance]
categorize grievance
alt
[Delay]
assignPriorityBasedOnSLA()
[Rejection]
linkRejectedApplication()
[Payment]
flagFinancialReview()
[Misconduct]
markHighPriority()
assignInvestigationQueue()
updateGrievanceStatus(Categorized)
categorization acknowledgement
investigate grievance
loadCaseHistory()
retrieveAuditLogs()


fetchAuditTrail()
checkSLACompliance()
alt
[SLA violated]
mandatoryEscalation()
notify supervisor
alt
[Can resolve at this level]
resolve grievance with remarks
submitResolution()
updateGrievanceStatus(Resolved)
logResolutionAudit()
resolution notification
[Escalation required]
escalate grievance
escalateGrievance()
updateGrievanceStatus(Escalated)
logEscalationAudit()
escalation alert
escalation update